Opakování goniometrických funkcí

1. Načrtněte grafy funkcí, určete jejich **periodu** a **obor hodnot**.

- $y = 2\sin(x)$
- $y = -\cos(x)$
- $y = \sin(2x)$
- $y = \cos(\frac{x}{2})$
- $y = \frac{1}{2}\sin(x)$
- $y = 3\cos(x)$
- $y = \sin(x) + 1$
- $y = \cos(x) - 2$
- $y = -2\sin(x)$
- $y = -3\cos(x)$
- $y = 2\sin(2x)$
- $y = -3\cos(\frac{x}{2})$
- $y = \sin\left(x - \frac{\pi}{2}\right)$
- $y = \cos\left(x + \frac{\pi}{4}\right)$
- $y = 2\sin(x) - 1$
- $y = -2\cos(x) + 2$
- $y = 3\sin\left(x - \frac{\pi}{2}\right)$
- $y = -\cos(2x) + 1$
- $y = \frac{1}{2}\sin(2x) + 1$
- $y = 2\cos(\frac{x}{2}) - 1$
- $y = 2\sin(2x - \pi) + 1$
- $y = -2\cos\left(\frac{x}{2} + \frac{\pi}{8}\right) - 1$
- $y = 3\sin\left(2x + \frac{\pi}{2}\right) - 2$
- $y = -\cos\left(\frac{x}{2} - \frac{\pi}{4}\right) + 2$
- $y = \frac{1}{2}\sin\left(2x - \frac{\pi}{2}\right) - 1$
- $y = -3\cos(2x + \pi) + 1$
- $y = 2\sin\left(\frac{x}{2} + \frac{\pi}{4}\right) - 2$
- $y = -2\cos\left(2x - \frac{\pi}{2}\right) + 1$
- $y = 3\sin\left(\frac{x}{2} - \frac{\pi}{8}\right) + 2$
- $y = \frac{1}{2}\cos\left(2x + \frac{\pi}{2}\right) - 2$

2. Určete **periodu** funkcí a rovnice jejich **vertikálních asymptot**.

Vzorec asymptot:

$$\boxed{tan : x = c + \frac{\frac{\pi}{2} + k\pi}{b}, \qquad k \in \mathbb{Z}}$$

$$\boxed{cot : x = c + \frac{k\pi}{b}, \qquad k \in \mathbb{Z}}$$

- $y = 2\tan\left(2x - \frac{\pi}{2}\right)$
- $y = -\cot\left(\frac{x}{2} + \frac{\pi}{4}\right)$
- $y = \tan\left(x - \frac{\pi}{4}\right) + 1$
- $y = -2\cot\left(x + \frac{\pi}{2}\right)$
- $y = 3\tan\left(2x - \frac{\pi}{2}\right) + 1$
- $y = 2\cot\left(\frac{x}{2} + \frac{\pi}{8}\right) - 1$
- $y = -\tan(2x + \pi) + 2$
- $y = -3\cot\left(2x - \frac{\pi}{2}\right) - 1$
- $y = 2\tan\left(\frac{x}{2} + \frac{\pi}{4}\right) + 1$
- $y = \cot\left(2x + \frac{\pi}{2}\right) - 2$
- $y = -2\tan\left(x - \frac{\pi}{4}\right) + 1$
- $y = 3\cot\left(x + \frac{\pi}{2}\right) - 1$
- $y = \frac{1}{2}\tan(2x - \pi) + 2$
- $y = -\frac{1}{2}\cot\left(\frac{x}{2} - \frac{\pi}{8}\right) + 1$
- $y = 2\tan\left(2x + \frac{\pi}{2}\right) - 2$
- $y = -2\cot(2x + \pi) + 2$
- $y = 3\tan\left(\frac{x}{2} - \frac{\pi}{4}\right) - 1$
- $y = -3\cot\left(\frac{x}{2} + \frac{\pi}{8}\right) + 2$
- $y = \tan\left(2x - \frac{\pi}{2}\right) - 1$
- $y = \cot\left(\frac{x}{2} - \frac{\pi}{4}\right) + 1$
- $y = -2\tan(2x + \pi) - 2$
- $y = 2\cot\left(\frac{x}{2} + \frac{\pi}{4}\right) - 1$
- $y = -3\tan\left(x + \frac{\pi}{4}\right) + 2$
- $y = 3\cot(2x - \pi) + 1$
- $y = \frac{1}{2}\tan\left(\frac{x}{2} - \frac{\pi}{8}\right) - 2$
- $y = -\frac{1}{2}\cot\left(2x + \frac{\pi}{2}\right) + 2$
- $y = 2\tan\left(x - \frac{\pi}{2}\right) - 1$
- $y = -2\cot\left(x + \frac{\pi}{4}\right) + 1$
- $y = 3\tan\left(2x + \frac{\pi}{2}\right) - 1$
- $y = -3\cot\left(\frac{x}{2} - \frac{\pi}{4}\right) - 2$